PRODUCT REQUIREMENTS DOCUMENT

Desa Works

A community Resource Management System for BUMDes to optimize local workforce allocation and track village project revenue.

Version	v1.0 - Draft
Date	May 10, 2026
Team	Aldi (24523023), Abdullah (24523229), Kemal (24523123), Dani (24523207)
Product Owner	Group 2
Client / Stakeholder	BUMDes Management
Status	Draft

How to use this template
You will add more sections in later sessions.
Blue underlined text = your group fills this in. Gray italic text = guidance; delete before submitting. Tables are pre-structured, add rows as needed, do not remove columns.

1. Problem Statement

1.1 Background & Context

Village economic projects managed by BUMDes (Badan Usaha Milik Desa) often rely on manual coordination for labor. Currently, there is no centralized database of residents' skills or availability, leading to underutilized human capital and inefficient project management. Without structured data, village leaders struggle to prove the economic impact of their initiatives to stakeholders.

1.2 Problem Statement

Format
[Affected group] cannot [do something] because [root cause], which results in [measurable negative impact].
Example: 'Citizens in urban areas cannot book vaccination appointments in advance because no digital scheduling system exists, producing unmanageable queues, missed doses, and no way for the Ministry to forecast clinic-level vaccine demand.'

BUMDes Management cannot efficiently manage and allocate local human resources to village economic activities because no structured digital system exists to track skills and participation, which results in inefficient project execution, missed revenue opportunities, and poor data-driven decision-making.

1.3 Who is Affected

- **BUMDes Director (Primary):** Struggles with manual worker assignments and lacks visibility into project progress.
- **Village Residents (Secondary):** Face unfair distribution of job opportunities due to the lack of a formal skills registry.
- **Village Head (Stakeholder):** Cannot accurately assess the economic contribution of village projects to the community.

2. Objectives

2.1 Business Objectives

#	Objective	Why it matters	Success indicator
1	Centralize workforce data	Enables faster matching of skills to project needs.	100% of project-active residents registered in the database.
2	Automate worker allocation	Reduces time spent on manual coordination.	50% reduction in time from project creation to worker

#	Objective	Why it matters	Success indicator
			assignment.
3	Track revenue per project	Ensures transparency and helps identify profitable ventures.	Monthly revenue reports generated automatically by the system.

2.2 User Objectives

Actor	What they need to accomplish	What stops them today
Resident	Register skills and get notified of jobs.	No platform to showcase their capabilities to village management.
Manager	Assign the right workers to specific tasks.	Reliance on memory or informal lists of who is "available."

3. Success Metrics

Metric	Baseline (now)	Target (3 months)	How it is measured
Workforce Registry Size	0 (Manual lists)	200+ Residents	Database count of RESIDENT table.
Assignment Speed	~3-5 Days	< 24 Hours	Timestamp difference between Project Creation and Assignment.
Reporting Accuracy	Manual/Incomplete	100% Financial Transparency	Matching system revenue logs with bank statements.

What counts as a real metric

'User satisfaction' is not a metric. 'Average time to complete vaccine registration, measured via server-side session logs' is. Before writing any metric, ask: can I measure this before launch and again after? If the answer is no, replace it.

4. Scope

4.1 In Scope & Out of Scope (MVP)

✅ IN Scope (MVP)	❌ OUT of Scope (v1)
Resident skills registration and	Automatic payroll/salary disbursements.

categorization.	
Project creation and workforce requirement definition.	Mobile App (MVP will be Web-based).
Skill-based worker recommendation engine.	Integration with national ID (Dukcapil) APIs.
Work progress recording and revenue tracking.	External vendor management.

4.2 Assumptions & Constraints

Type	Description
Assumption	Residents have access to devices (mobile or PC) to input data or can use a kiosk at the BUMDes office.
Constraint	System must run on local servers (XAMPP/MySQL) as per initial technical requirements.
Constraint	Legal Compliance: Data management must comply with Personal Data Protection Law (UU PDP) No. 27 of 2022.

4.3 Data Protection & Encryption Requirements

To comply with Indonesia's Personal Data Protection Law (UU No. 27 of 2022), the system shall implement:

- **Data in transit**: All communication between the user's browser and the server must use HTTPS (TLS 1.2 or higher).
- **Data at rest**: Personal data (name, contact information, skills) must be encrypted in the database using AES-256. MySQL's built-in InnoDB tablespace encryption will be enabled.
- **Passwords**: Stored using one-way hashing with bcrypt (cost factor ≥ 10) or argon2.
- **Encryption keys**: Must not be hard-coded in source code; keys will be stored in environment variables or a local secrets manager, separated from the data.
- **Backups**: All database backup files must also be encrypted.
- **User consent**: Residents must explicitly agree to the platform's Terms of Service and Privacy Policy during registration, providing opt-in consent for personal data processing as required by the PDP Law.

5. Functional Requirements

Before you submit each FR, check four things
Unambiguous: two engineers reading it independently would build the same thing.
Testable: you can write a pass/fail test case for it right now.
Feasible: it can be built within this project's constraints.
Prioritized: you have a MoSCoW label and you would defend it in a scope-cut conversation.

5.1 FR Table: Resident

FR ID	Actor	The system shall...	Condition / Trigger	Priority	MoSCoW
FR-R 01	Resident	Allow users to input skills and availability.	When the resident creates or updates their profile.	High	M
FR-R 02	Resident	The system shall display a notification on the resident's dashboard and, if an email is provided, send an email.	When Manager confirms the assignment.	Medium	S
FR-R 03	Resident	Update personal availability status	When resident becomes unavailable or available	Medium	S

5.2 FR Table : Manager

FR ID	Actor	The system shall...	Condition / Trigger	Priority	MoSCoW
FR-M 01	Manager	Create projects with specific skill requirements.	Project initiation	High	M
FR-M 02	Manager	Recommend workers based on skill matching. The system returns a list of residents whose skills match the	After project requirements are set	High	M

FR ID	Actor	The system shall…	Condition / Trigger	Priority	MoSCoW
		minimum required skill and are available			
FR-M 03	Mana ger	Generate project performance and revenue reports	Dashboard request	High	M
FR-M 04	Mana ger	Notify assigned workers regarding project assignments	After assignment confirmation	Medium	S
FR-M 05	Mana ger	View active project progress dashboard	When manager accesses dashboard	High	M
FR-M 06	Mana ger	Filter workers based on skills and availability	During workforce assignment process	Medium	S

MoSCoW reference
M Must Have: the product does not ship without this.
S Should Have: significant value, expected in the next sprint or release.
C Could Have: nice to have; only included when higher-priority items are done.
W Won't Have (this time): deferred. Write it here so it cannot silently re-enter scope.

6. User Workflows

6.1 Workflow: Project Assignment & Workforce Allocation

Actor	BUMDes Manager
Goal	Successfully assign qualified residents to a new village project.
FRs covered	FR-M01, FR-M02, FR-M04, FR-M06

Ideal Path

#	Step description
1	Manager initiates a project (e.g., "Mushroom Farm Expansion").
2	Manager defines required skills (e.g., "Agriculture," "Basic Accounting").
3	System analyzes RESIDENT_SKILL table and recommends available workers.
4	Manager reviews and confirms the assignment list.
5	System updates ASSIGNMENT status and triggers worker notification.

Decision Points

Decision Point	YES / Success path	NO / Error path
Are there enough residents with the required skills?	System shows recommended list	Alert manager: “Insufficient workers on skill X. Broaden the requirements or create a job posting.”
Does manager Confirm the final assignment list?	System saves assignment, triggers notifications	Return to recommended list for editing

Edge Cases

Edge Case	What the system must do
A recommended resident is already assigned to another project with overlapping dates	Flag resident as “conflict” and exclude from auto-selection, showing warning
Manager changes requirements after recommendations are shown	System recalculates recommendations automatically, clearing previous selection.
Resident’s availability changes after assignment (for example, they leave village)	Allow manager to unassign and re-search, mark previous assignment as void

6.2 Workflow: Project Monitoring & Revenue Tracking

Actor	Worker / Manager
Goal	Record project progress, monitor worker contributions, and generate financial analytics for managerial decision-making.
FRs covered	FR-M03, FR-M05, FR-R03

Ideal Path

#	Step description
1	Worker accesses assigned project tasks from dashboard.
2	Worker updates work progress and contribution details.
3	System validates and stores submitted progress records.
4	Manager reviews project progress through dashboard analytics.
5	Manager records final project results and generated revenue.
6	System generates updated performance and revenue reports automatically.

Decision Points

Decision Point	YES / Success path	NO / Error path
Is the submitted progress data complete?	System saves progress update successfully	System requests missing or invalid information
Has the project been completed?	System enables final revenue recording and report generation	System keeps project status active for further updates

Edge Cases

Edge Case	What the system must do
Worker submits duplicate progress update	System detects duplicate submission and asks user to confirm or edit
Revenue exceeds expected budget range	System flags abnormal financial input and alerts manager
Internet disconnects during submission	System temporarily saves draft and allows retry submission

6.3 Workflow: Workforce Skills Registration

Actor	Resident / Manager
Goal	Register resident skills and availability into the workforce database for future project allocation.
FRs covered	FR-R01, FR-R03

Ideal Path

#	Step description
1	Resident accesses the registration form through the system dashboard.
2	Resident inputs personal information, skills, experience, and availability status.
3	System validates submitted information for completeness and formatting.
4	System stores resident profile and skill data in the workforce database.
5	Manager reviews newly registered profiles if verification is required.
6	System updates the available workforce list for future project assignments.

Decision Points

Decision Point	YES / Success path	NO / Error path
Is all required registration information complete?	System accepts registration and stores data successfully	System highlights missing or invalid fields for correction
Does the manager approve the submitted profile?	Resident profile becomes active in workforce database	System marks profile as pending revision or rejected

Edge Cases

Edge Case	What the system must do
Resident submits duplicate registration data	System detects duplicate records and prompts resident to update existing profile instead
Resident enters unsupported or unclear skill descriptions	System suggests predefined skill categories or requests clarification
Resident changes availability after approval	System updates workforce database immediately and reflects changes in assignment recommendations

Revision History

Update this table each time you change the document. Version, date, author, one-line summary of what changed.

Version	Date	Author	Changes
v0.1	08 May 2026	All	Initial draft
v0.2	[Date]	[Name]	[What changed]